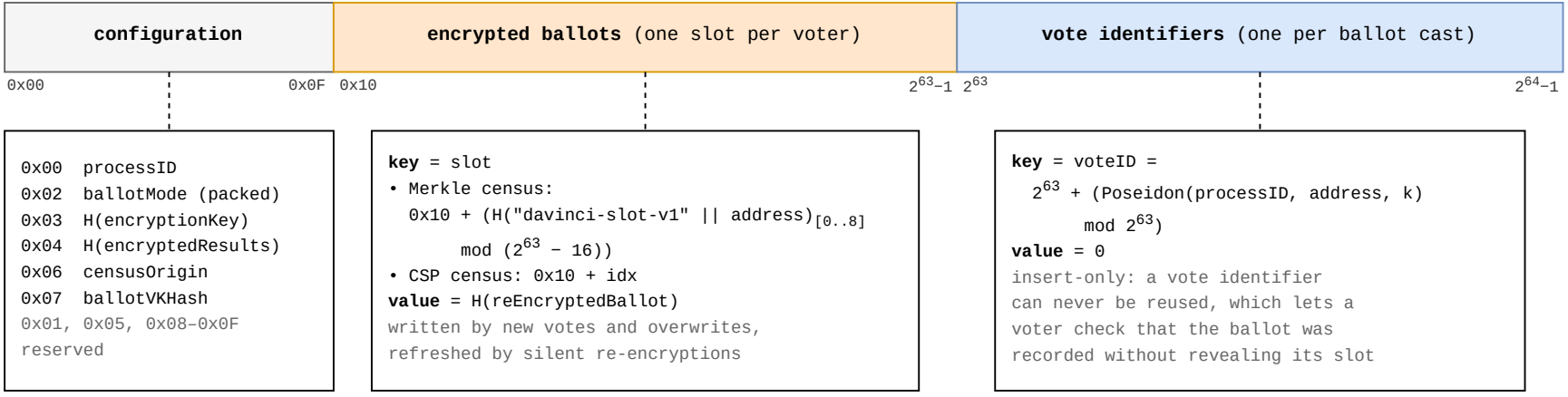
state tree: sparse Merkle tree, depth 64, SHA-256, 64-bit keys; its root is the on-chain **stateRoot**



H = SHA-256 (leaf value read as a big-endian integer). Leaf hash: $H(key_{LE8} \,||\, value_{LE32} \,||\, 0x01)$; internal node: H(left || right). Genesis: the registry computes the root of the six configuration leaves at creation, with the accumulator set to the encryption of zero (all identity points).